



Digital Farmer Assistance System

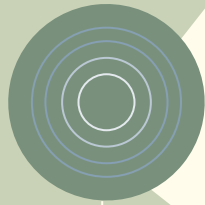


01

Problem Statement

Problem Statement

Core Issue



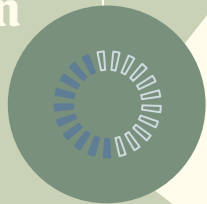
Farmers currently lack timely and reliable access to localized crop advisory, real-time market prices, and critical weather information.

Consequences

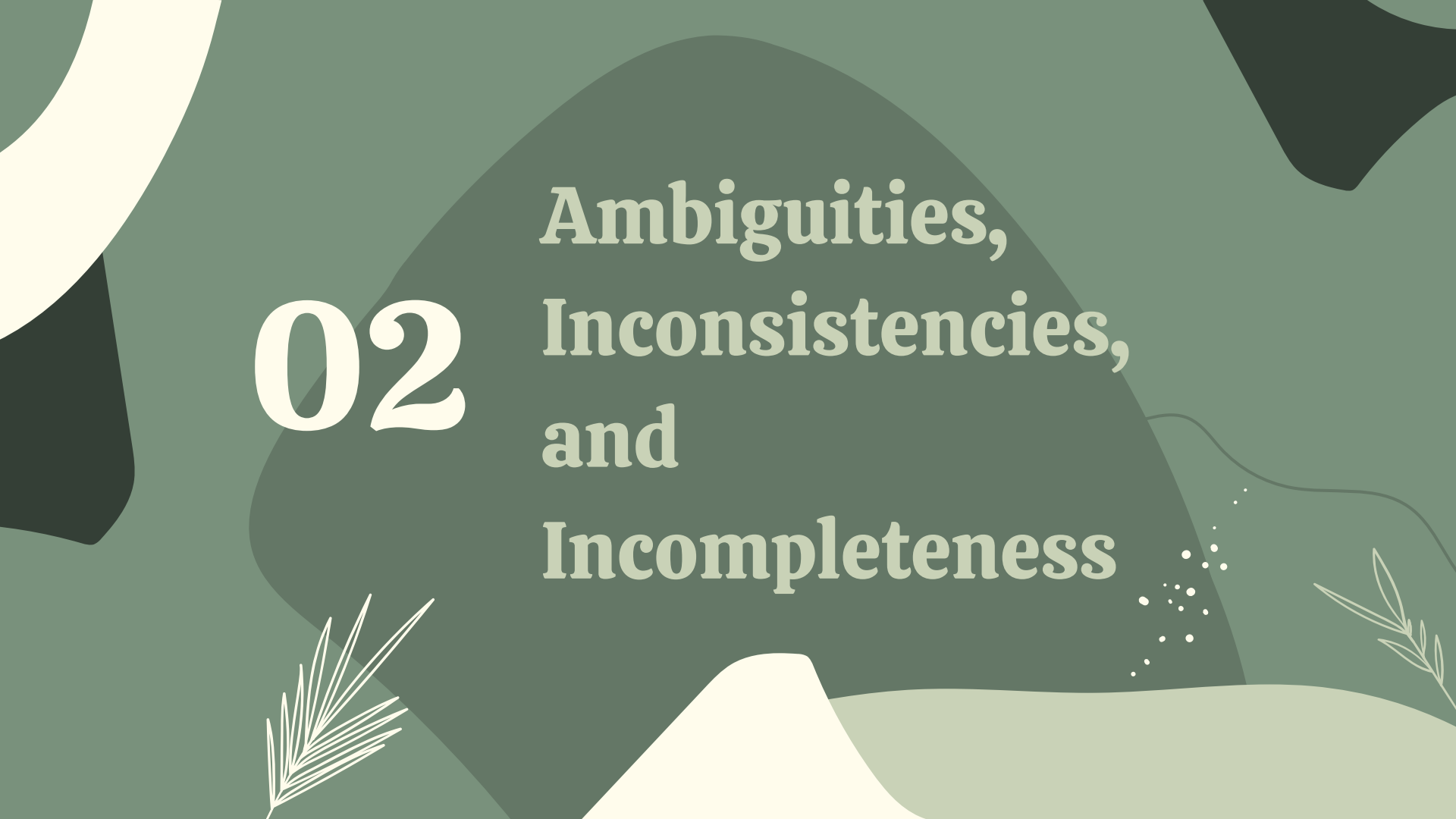


This information gap results in inefficient farming practices, reduced crop yields, and financial vulnerability due to market volatility and unpredictable weather patterns.

Proposed Solution



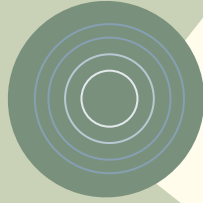
A digital platform to bridge the communication gap between agricultural experts, market data and farming community



02

Ambiguities,
Inconsistencies,
and
Incompleteness

Ambiguities



- "Timely access" is not defined; does it mean real-time, hourly, or daily updates?
- "Market prices" does not specify the geographical range (local mandis vs. national markets).

Inconsistencies

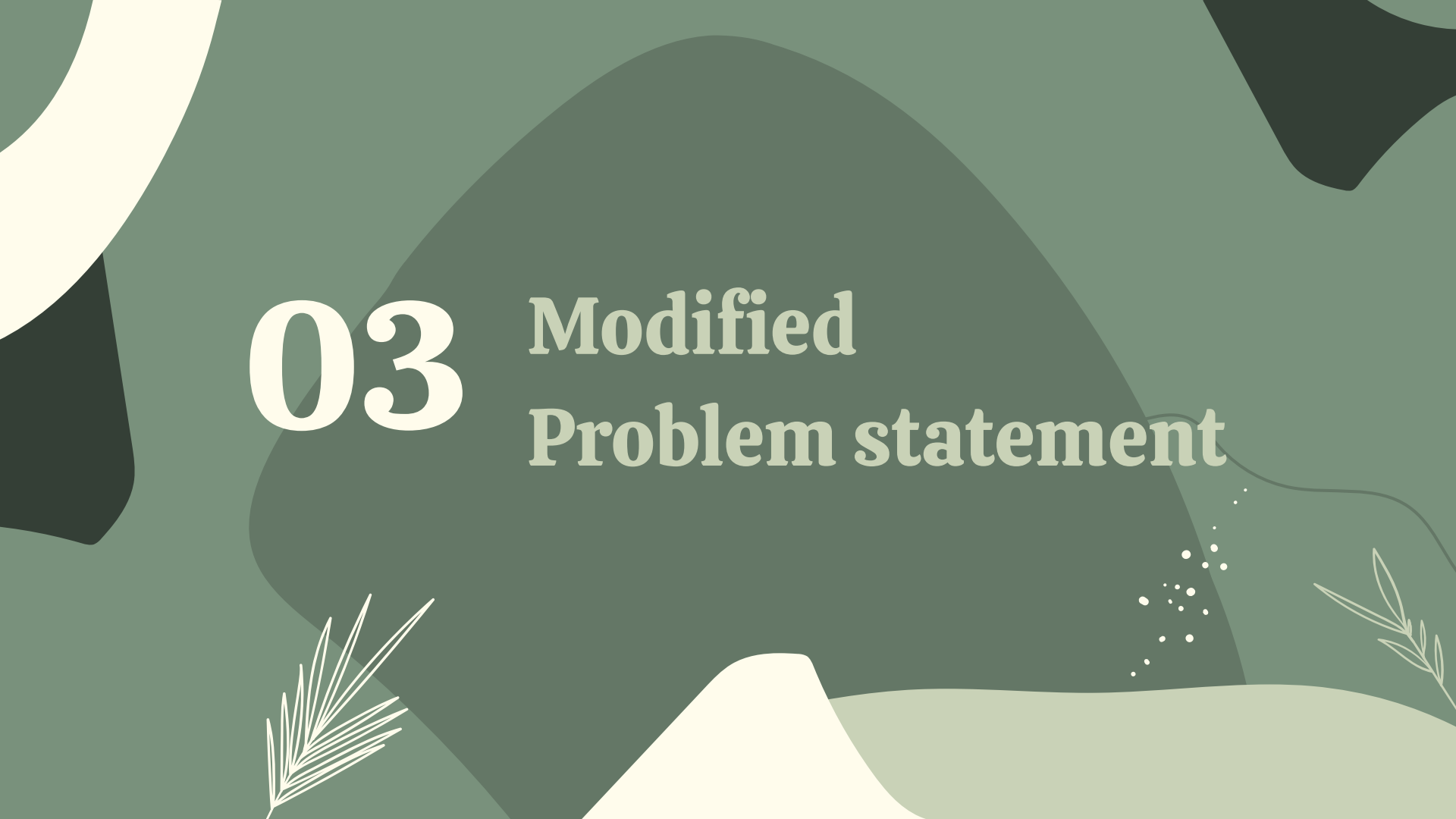


- The initial scope mentions providing "weather alerts" but doesn't clarify if these are automated or manually pushed by admins.
- Mention of "low bandwidth" requirements vs. "web/mobile-based access" (complex web apps often struggle on low-speed rural networks).

Incompleteness



- No mention of how data is sourced or validated (data integrity).
- Missing details on offline functionality for farmers in areas with intermittent connectivity
- Lack of specific user authentication protocols for data privacy.



03

Modified Problem statement

Modified problem statement

To design and develop a Digital Farmer Assistance System that centralizes agricultural data to provide:

- Real-time synchronization of regional market prices and weather forecasts.
- Validated, season-specific crop advisories through a multilingual interface

Goal: To eliminate information asymmetry, allowing farmers to make data-driven decisions that enhance productivity and financial sustainability in a low-bandwidth environment.



04 Functional Requirements

Functional Requirements

01

User management

The system shall allow farmers to register, create profiles, and select preferred crops/regions.

02

Information Retrieval

The system shall display daily updated market prices and localized 5-day weather forecasts.

03

Advisory Engine

The system shall provide crop-specific guidance based on the current growth stage and season.

04

Alert system

The system shall push automated SMS/App notifications for extreme weather warnings.

05

Admin control

The system shall provide a dashboard for administrators to update prices and advisory content.

06

Interaction

The system shall include a feedback/query module for farmers to interact with agricultural experts.



05 Non-Functional Requirements

Non Functional Requirements

01

Usability

The interface must follow a "Mobile-First" design with high-contrast icons for users with varying digital literacy.

02

Availability

The system shall maintain 99.9% uptime (24/7 availability).

03

Performance

The platform must load essential text data within 2 seconds on 2G/Low-bandwidth connections.

04

Accessibility

The system shall support at least three regional languages in addition to English.

05

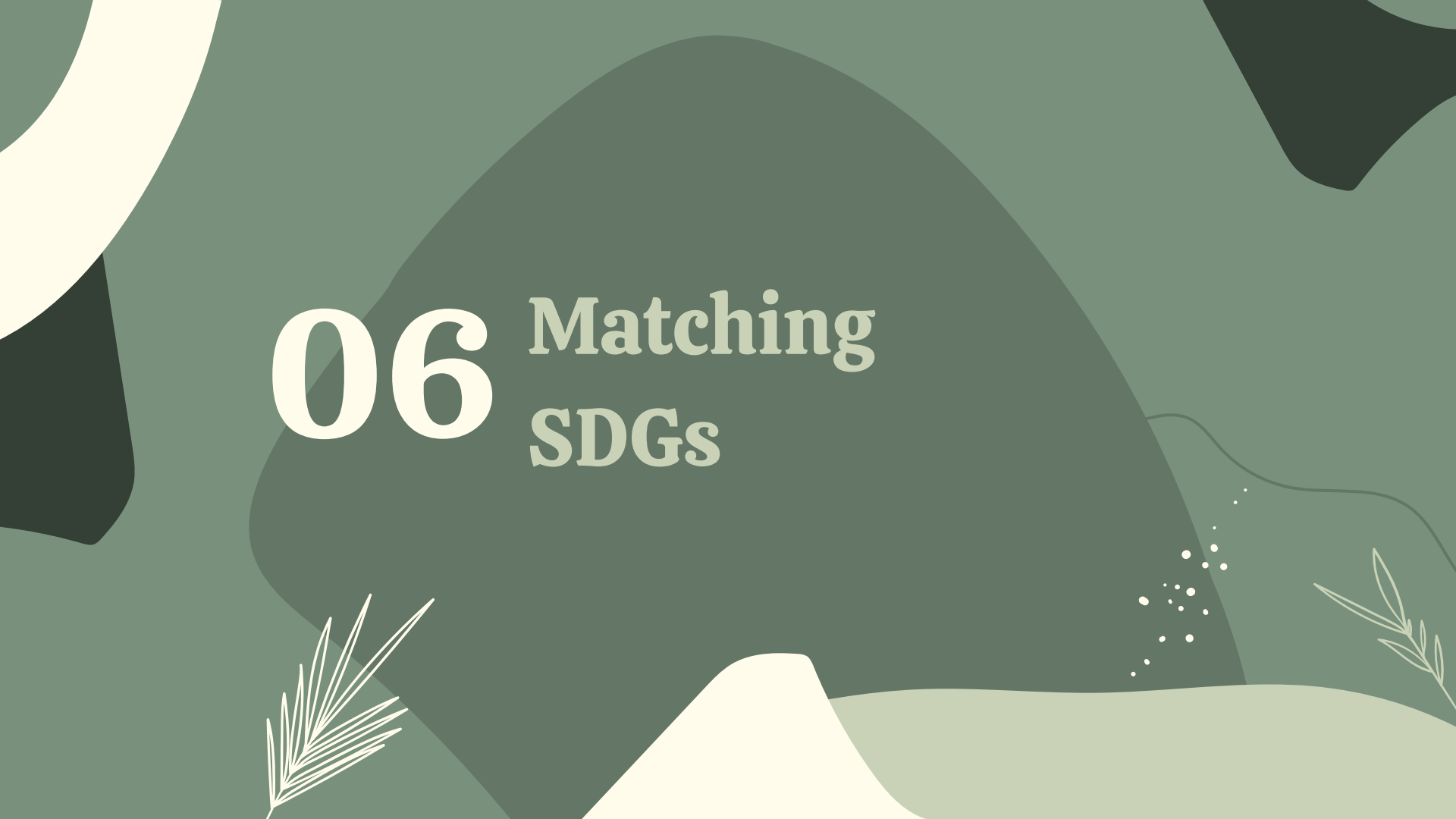
Scalability

The architecture must support a concurrent user increase of 20% annually without performance degradation.

06

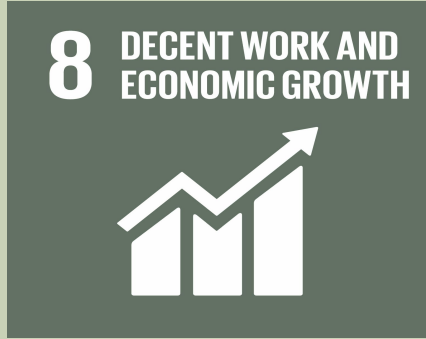
Security

All user data and profile information must be encrypted and accessible only through secure logins.



06 Matching SDGs

Following SDG's:



SDG 8 – Decent Work and Economic Growth

Increasing the income potential of small-scale farmers by providing direct access to market trends.



SDG 9 – Industry, Innovation, and Infrastructure

Utilizing digital infrastructure to modernize traditional agricultural practices and enhance rural connectivity.



07 Feasibility Analysis



Technical feasibility

- The system is technically feasible as it can be implemented using existing web and mobile technologies.
 - Crop advisory, market prices, and weather information can be obtained using reliable third-party APIs.
 - Multilingual support and user authentication can be implemented using standard frameworks.
 - Since the system does not require hardware integration or advanced AI models, the technical complexity remains manageable.
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Economic feasibility

- The project is economically feasible because it uses open-source tools and commonly available cloud infrastructure, which minimizes development and deployment costs.
 - Farmers require only basic smartphones or computers with internet access.
 - The benefits of improved decision-making, better crop planning, and increased income justify the cost of system development and maintenance.
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Operational feasibility

- The system is operationally feasible as it is designed to be simple and user-friendly for rural users.
 - Farmers can easily access crop advisories, market prices, and weather information.
 - Administrators can manage content through an admin module.
 - Minimal training is required, making the system easy to adopt in real-world scenarios.
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Schedule feasibility

- The project can be completed within the academic time frame.
- The modular structure allows independent development of features such as user management, advisory services, and weather updates.
- This ensures timely completion and future extensibility.

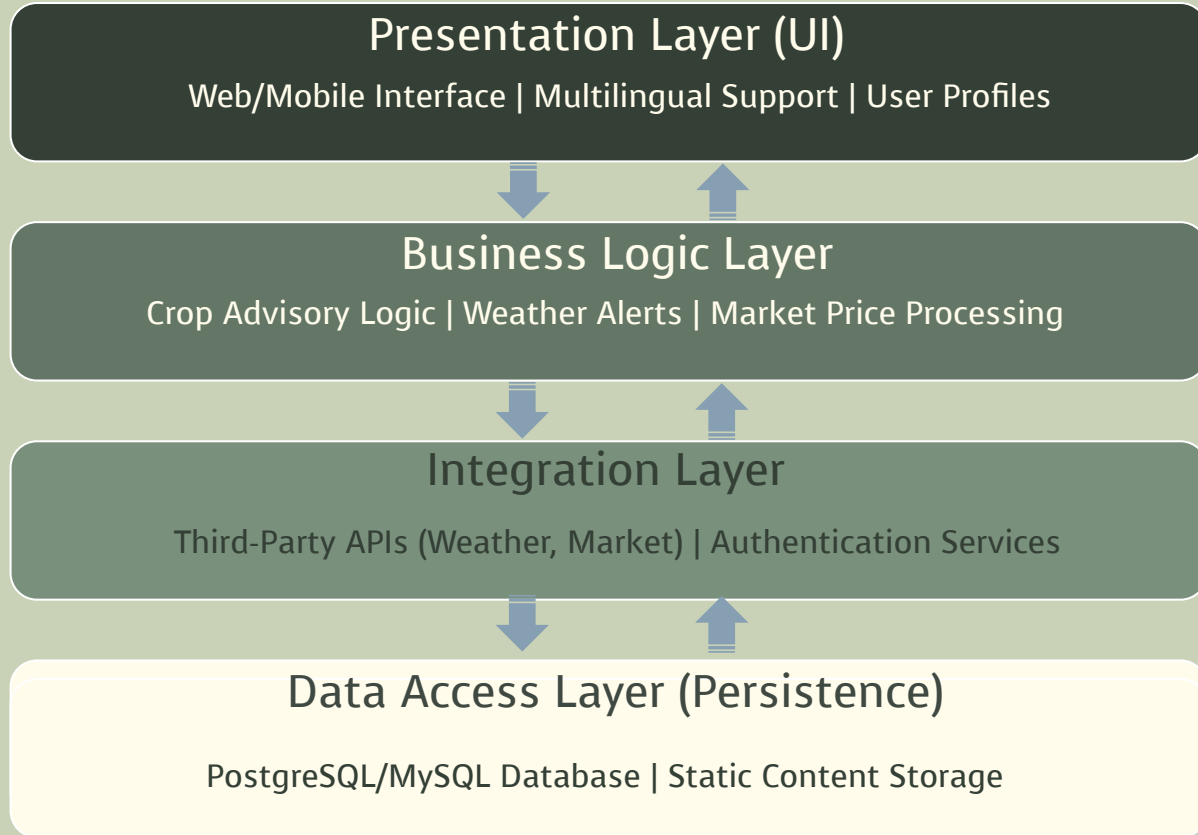
Based on the analysis, the Digital Farmer Assistance System is technically, economically, operationally, and schedule feasible.





08 Architectural Design

Layered Architecture Design





Presentation Layer (UI)

- **Handles the user interface for Farmers and Admins.**
 - **Implements Multilingual Support to ensure accessibility for rural users.**
 - **Responsive web/mobile views designed for low-bandwidth environments.**
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Business Logic Layer (Service Layer)

- **Advisory Engine:** Processes season and crop type to provide tailored guidance.
 - **Alert Manager:** Logic for triggering weather notifications and market price updates.
 - **User Management:** Handles registration and profile validation.
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Integration Layer

- **Facilitates communication with Third-Party APIs for real-time weather and market data.**
 - **Abstracts the external data sources so the core system remains independent of API changes.**
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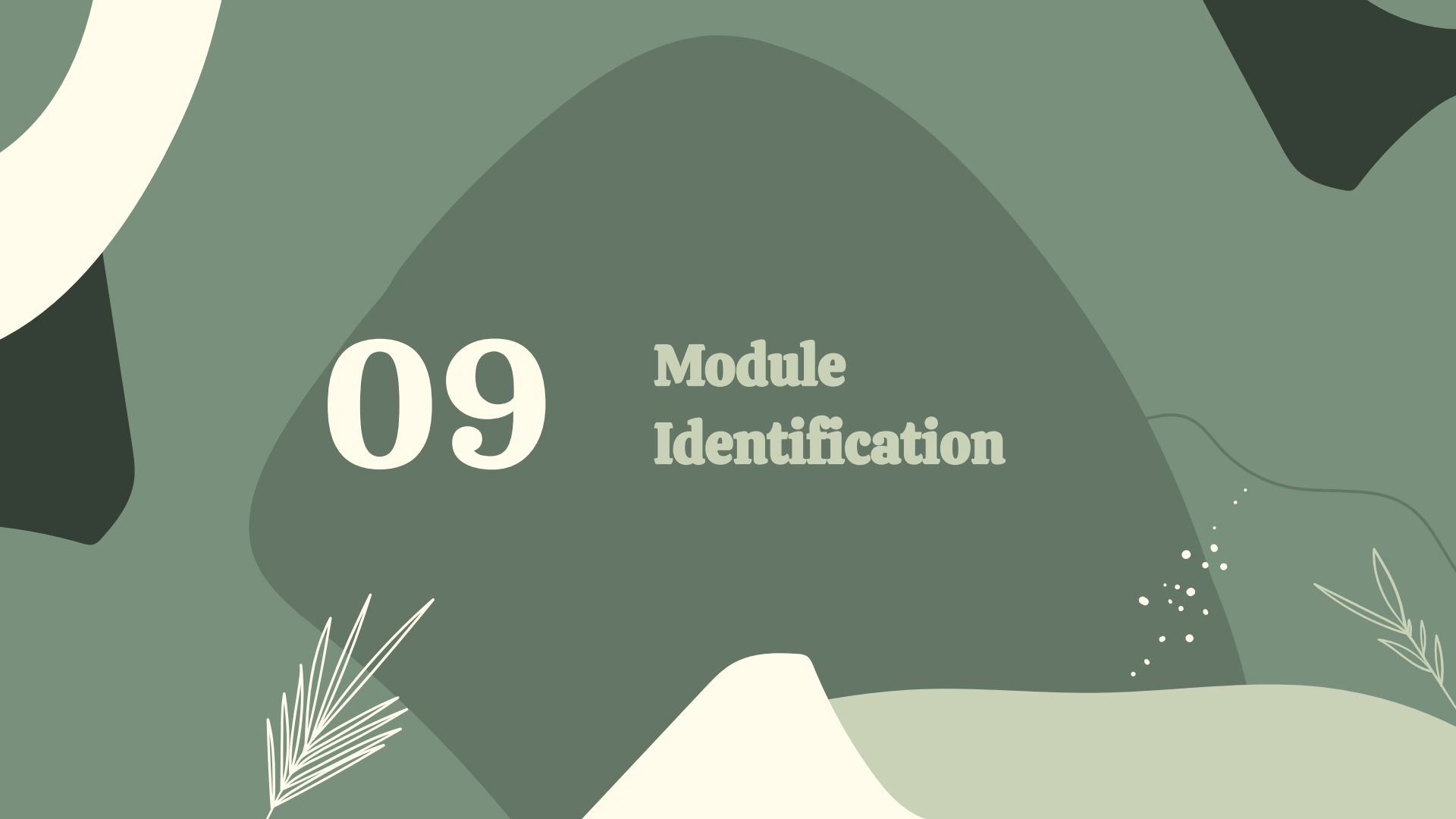
Data Access Layer (Persistence)



- **Manages the Backend Database System.**
 - **Stores farmer profiles, historical price trends, and localized advisory content.**
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Design Justification

- **Scalability:** By using a layered approach, we can later transition to a Microservices Architecture or a Cloud-based "Everything as a Service" (XaaS) model if the user base grows. . . .
- **Maintainability:** Modularity makes "Corrective" and "Perfective" maintenance easier because changes in one layer don't break the other.
- **Reliability:** Decoupling the data fetching from the display ensures that if an external API is slow, the app remains responsive.



09

**Module
Identification**



Presentation Layer (UI)

User Interface Module: Manages the "Mobile-First" responsive views and high-contrast icons for rural accessibility.



Multilingual Support Module: Provides the translation logic to support regional languages plus English.





Business Logic Layer (Service Layer)

User Management Module: Handles the logic for registration, secure login via encrypted profiles, and role-based access for Farmers and Admins.

Advisory Engine Module: Processes specific crop types and seasonal stages to generate tailored agricultural guidance.



Alert Manager Module: Contains the logic for triggering notifications based on extreme weather warnings or significant market price fluctuations.



Integration Layer

Market Price Integration Module: Fetches real-time pricing from external agricultural APIs and passes it to the Business Layer.



Weather Service Module: Maintains the connection to third-party weather providers for 5-day forecasts and localized data.





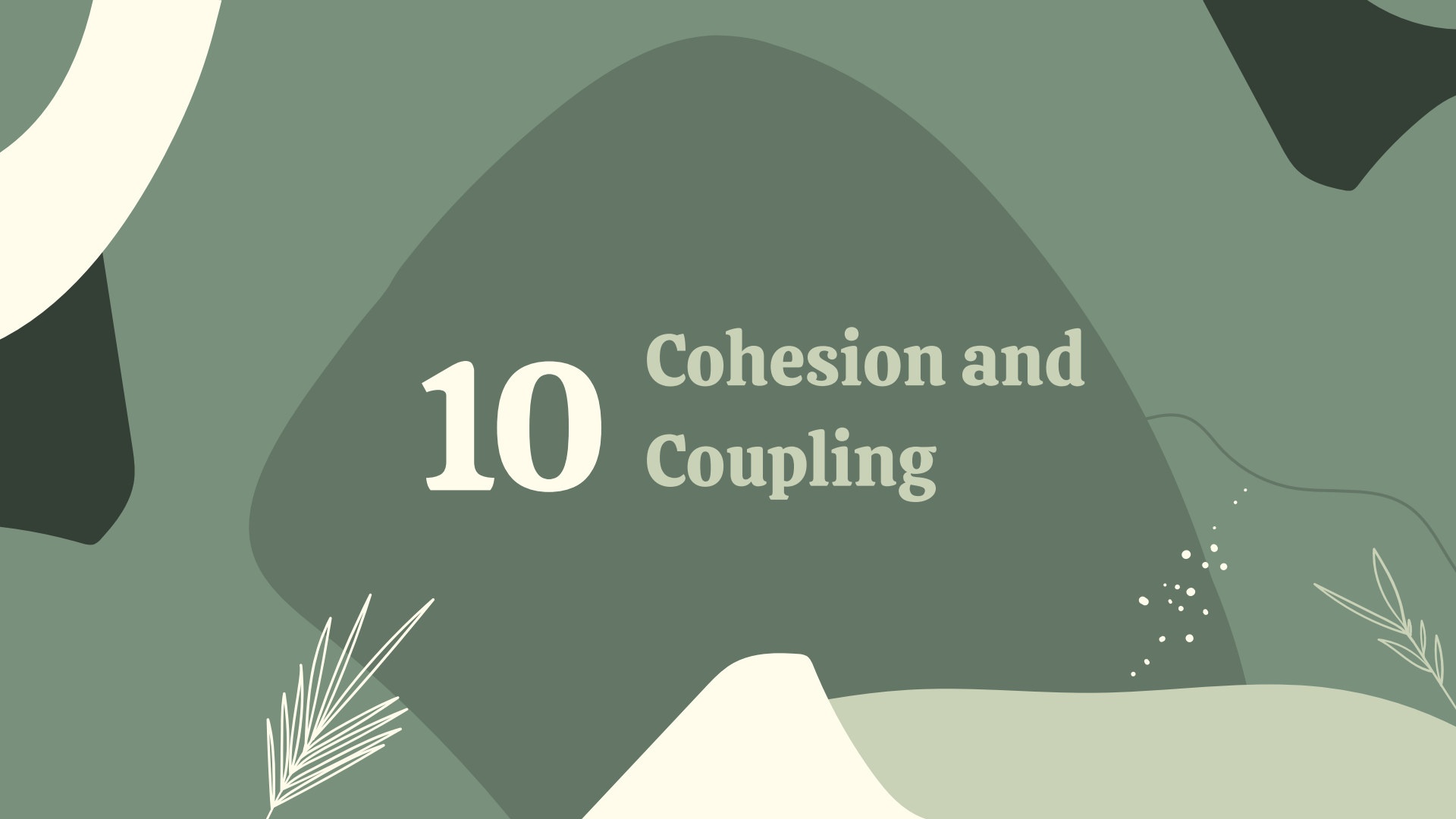
Data Access Layer (Persistence)

Data Persistence Module: Manages the PostgreSQL/MySQL database, ensuring data consistency for farmer profiles and historical price trends.



Content Management Module: Specifically handles the storage and retrieval of static advisory content uploaded by administrators.





10 Cohesion and Coupling



Cohesion Analysis

- Each module performs a single well-defined function, ensuring high cohesion.

EXAMPLES

- User Management Module focuses only on authentication and profiles.
 - Crop Advisory Module handles advisory logic without UI or data storage concerns.
 - Weather & Alert Module is responsible only for notifications and alerts.
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Benefits of High Cohesion

- Easier maintenance and debugging.
 - Improved code readability and reusability.
 - Modules can be independently enhanced or replaced.
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Coupling Analysis

- The system follows low coupling through a layered architecture.

Modules communicate through:

- Defined interfaces
- Service calls rather than direct data access

Examples

- UI layer interacts with the Business Logic Layer, not directly with the database.
 - Business Logic accesses external data only via the Integration Module.
 - Changes in third-party APIs do not affect core system modules.
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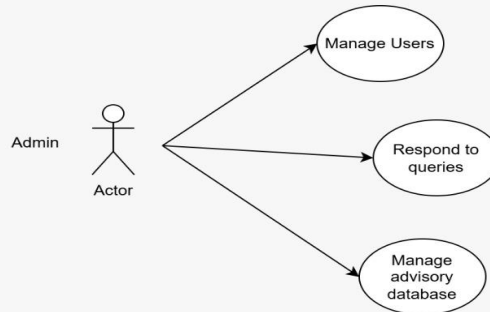
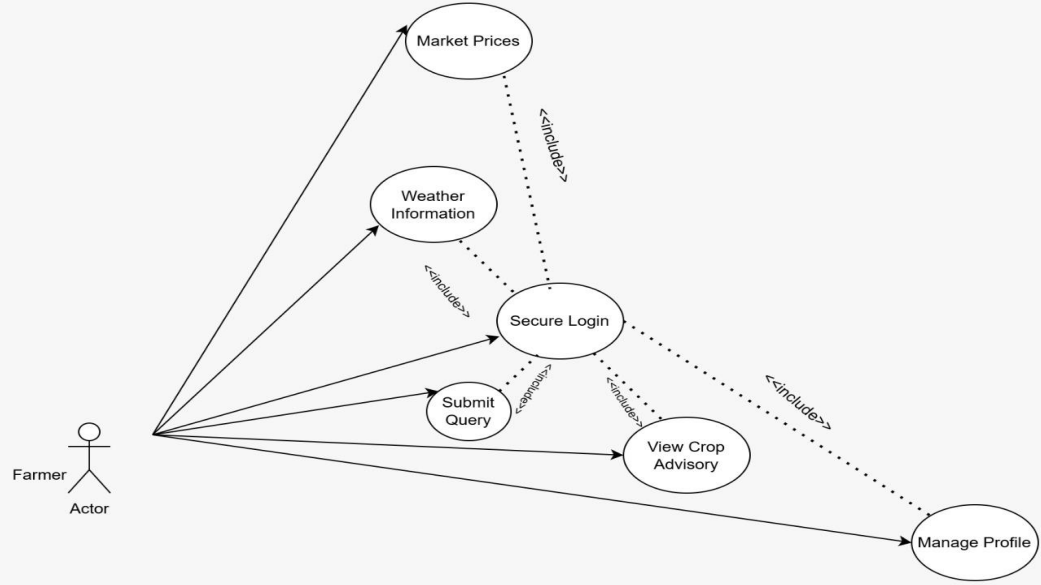
Benefits of Low Coupling

- Reduces impact of changes across modules.
 - Enhances system scalability and flexibility.
 - Supports future migration to microservices or cloud-based architecture.
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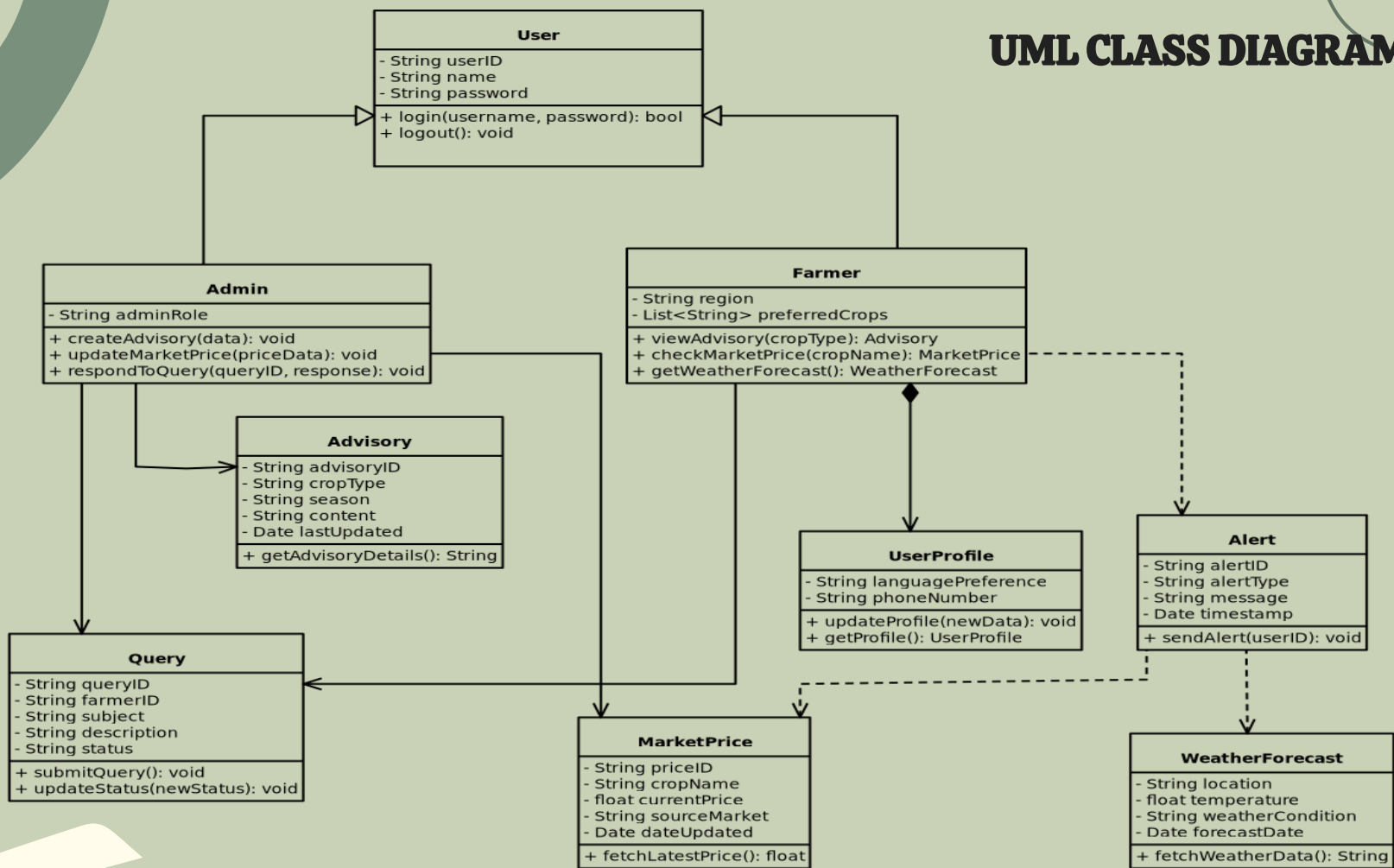


11 UML Diagrams

UML USE CASE DIAGRAM



UML CLASS DIAGRAM





END

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